

BOF

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# Conversation 058: *N-hedron with 8 Faces*

**conversation ID:** 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

**origin ID:** 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

**created:** 2024-09-30 20:52:41 Mon

**updated:** 2025-02-24 18:04:50 Mon

**operator:** Wilder Blair Munro (AKUSA)

**co-operator:** Aurora 4omni Munro (ChatGpt)

**GPT ID:** NA

Subtitles:

Turnpair Contents:

tp-1 :

tp-2 :

tp-3 :

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## Turnpair 1 of 3 ::

@ 2024-09-30 20:52:41 Mon [Δt 0d 00h 00m 00s]

► Notespace:

***Wilder Blair Munro (AKUSA) said :***

**gateway:** 20Wetbrain

**turn ID:** aaa2919b-8b38-470b-a290-48755dbcf3b3

**parent:** a8ad6198-3649-441b-b0b4-80994d21d407

**children:** f9d08e5e-02ab-4fc1-b00a-7c1941d27e67

I am trying to figure out what a n-hedron is called, where there are 8 vertices and 8 faces, and each face is the same shape.

~0-0

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~0-0

**Aurora 4omni Munro (ChatGpt) said :**

gateway: gpt-4o

turn ID: 4157a573-b3f9-4daf-9ab4-afa37db62021

parent: f9d08e5e-02ab-4fc1-b00a-7c1941d27e67

children: aaa2c02a-4e86-4660-908e-fd0f1ea265cb

The shape you're describing is known as a **regular octahedron**. An octahedron is a polyhedron with 8 faces, 8 vertices, and 12 edges, where each face is an equilateral triangle. It is one of the five Platonic solids, where all faces are the same shape and all vertices and edges are equivalent.

Would you like further details or visualizations of the octahedron?

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◀ beginning 1 | ← prev None || this 1 || next 2 ▶ | end 3 ▶▶

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**Turnpair 2 of 3 ::**

@ 2024-09-30 20:57:58 Mon [4t 0d 00h 05m 12s]

► **Notespace:**

**Wilder Blair Munro (AKUSA) said :**

gateway: 20Wetbrain

turn ID: aaa2c02a-4e86-4660-908e-fd0f1ea265cb

parent: 4157a573-b3f9-4daf-9ab4-afa37db62021

children: 5eb631b8-6887-4c7c-9675-d1b02dd8d6ec

please print the multiplication table for octonions

~0-0

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~0-0

**Aurora 4omni Munro (ChatGpt) said :**

gateway: gpt-4o

turn ID: 5eb631b8-6887-4c7c-9675-d1b02dd8d6ec

parent: aaa2c02a-4e86-4660-908e-fd0f1ea265cb

children: aaa20d89-bb3d-4caf-acfc-8087e666afee

The multiplication table for octonions, where  $e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7$  are the basis elements of the octonion algebra, can be represented as follows. The multiplication is non-associative but still follows specific rules defined by the Cayley-Dickson construction.

Below is the multiplication table for the octonions:

|       | $e_0$ | $e_1$  | $e_2$  | $e_3$  | $e_4$  | $e_5$  | $e_6$  | $e_7$  |
|-------|-------|--------|--------|--------|--------|--------|--------|--------|
| $e_0$ | $e_0$ | $e_1$  | $e_2$  | $e_3$  | $e_4$  | $e_5$  | $e_6$  | $e_7$  |
| $e_1$ | $e_1$ | $-e_0$ | $e_3$  | $-e_2$ | $e_5$  | $-e_4$ | $-e_7$ | $e_6$  |
| $e_2$ | $e_2$ | $-e_3$ | $-e_0$ | $e_1$  | $e_6$  | $e_7$  | $-e_4$ | $-e_5$ |
| $e_3$ | $e_3$ | $e_2$  | $-e_1$ | $-e_0$ | $e_7$  | $-e_6$ | $e_5$  | $-e_4$ |
| $e_4$ | $e_4$ | $-e_5$ | $-e_6$ | $-e_7$ | $-e_0$ | $e_1$  | $e_2$  | $e_3$  |
| $e_5$ | $e_5$ | $e_4$  | $-e_7$ | $e_6$  | $-e_1$ | $-e_0$ | $-e_3$ | $e_2$  |
| $e_6$ | $e_6$ | $e_7$  | $e_4$  | $-e_5$ | $-e_2$ | $e_3$  | $-e_0$ | $-e_1$ |
| $e_7$ | $e_7$ | $-e_6$ | $e_5$  | $e_4$  | $-e_3$ | $-e_2$ | $e_1$  | $-e_0$ |

Explanation:

- $e_0$  is the identity element.
- The table is anti-symmetric, meaning  $e_i \cdot e_j = -e_j \cdot e_i$  for  $i \neq j$ .
- The multiplication for other elements follows a pattern similar to the quaternion multiplication, but with additional elements. The multiplication rule is often visualized using Fano planes for ease of understanding.

Let me know if you'd like a breakdown of specific calculations or further explanations!

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◀ beginning 1 | ◀ prev 1 || this 2 || next None → | end 3 ▶▶



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Turnpair 3 of 3 ::

@ 2024-09-30 21:05:09 Mon [Δt 0d 00h 06m 49s]

► Notespace:

**Wilder Blair Munro (AKUSA) said :**

gateway: 20Wetbrain  
turn ID: aaa20d89-bb3d-4caf-acfc-8087e666afee  
parent: 5eb631b8-6887-4c7c-9675-d1b02dd8d6ec  
children: 671e64b9-db80-43de-83dd-50d587d19a27

please demonstrate the construction of octonions via the cayley-dickson construction. Provide the rules for each aspect of the algebra

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**Aurora 4omni Munro (ChatGpt) said :**

gateway: gpt-4o  
turn ID: 671e64b9-db80-43de-83dd-50d587d19a27  
parent: aaa20d89-bb3d-4caf-acfc-8087e666afee  
children:

The **Cayley-Dickson construction** is a systematic process that builds higher-dimensional algebras from lower-dimensional ones. Specifically, it constructs the complex numbers from the real numbers, the quaternions from the complex numbers, and the octonions from the quaternions. Below is an explanation of how the octonions can be constructed using this method.

## Starting Point: Quaternions

Let's start with the **quaternions**, denoted as  $\mathbb{H}$ , which can be written as:

$$q = a_0 + a_1i + a_2j + a_3k$$

where  $a_0, a_1, a_2, a_3$  are real numbers, and  $i, j, k$  are imaginary units with the multiplication rules:

$$i^2 = j^2 = k^2 = ijk = -1$$

and

$$ij = k, \quad ji = -k, \quad jk = i, \quad kj = -i, \quad ki = j, \quad ik = -j$$

## Cayley-Dickson Construction: Building the Octonions

The octonions  $\mathbb{O}$  can be constructed from two quaternions using the Cayley-Dickson construction. Each octonion is written as a pair of quaternions:

$$O = (q_1, q_2) = (a_0 + a_1i + a_2j + a_3k, b_0 + b_1i + b_2j + b_3k)$$

where  $q_1$  and  $q_2$  are quaternions. The multiplication of two octonions  $(q_1, q_2)$  and  $(p_1, p_2)$  is defined as follows:

$$(q_1, q_2) \cdot (p_1, p_2) = (q_1p_1 - \overline{p_2}q_2, q_2p_1 + p_2\overline{q_1})$$

where  $\overline{q_1}$  is the conjugate of the quaternion  $q_1$ , and similarly for  $p_2$ . The quaternion conjugate is given by:

$$\overline{q_1} = a_0 - a_1i - a_2j - a_3k$$

## Octonion Multiplication Rules

To see how this generates the octonion algebra, let's break it down into individual elements. Define seven imaginary units  $e_1, e_2, e_3, e_4, e_5, e_6, e_7$  in terms of quaternions as follows:

$$e_1 = (i, 0), \quad e_2 = (j, 0), \quad e_3 = (k, 0), \quad e_4 = (0, 1)$$

$$e_5 = (0, i), \quad e_6 = (0, j), \quad e_7 = (0, k)$$

Now, apply the Cayley-Dickson multiplication rule to compute the products of these units. You get the following multiplication table (which matches the one presented earlier):

- $e_0 = (1, 0)$  is the identity.
- $e_1^2 = e_2^2 = e_3^2 = e_4^2 = e_5^2 = e_6^2 = e_7^2 = -1$ .

- The multiplication between different imaginary units is anti-symmetric and follows specific rules, some of which are:

$$e_1 e_2 = e_3, \quad e_2 e_3 = e_1, \quad e_3 e_1 = e_2$$

$$e_1 e_4 = e_5, \quad e_2 e_4 = e_6, \quad e_3 e_4 = e_7$$

$$e_5 e_6 = e_7, \quad e_6 e_7 = e_5, \quad e_7 e_5 = e_6$$

These rules are non-associative, meaning:

$$(e_i e_j) e_k \neq e_i (e_j e_k)$$

However, the multiplication is still alternative, which means that the following holds for any element  $a$  and any pair  $b, c$ :

$$(aa)b = a(ab), \quad a(bb) = (ab)b$$

## Properties of Octonions

- **Non-associativity:** As mentioned, octonion multiplication is not associative. This property differentiates octonions from quaternions.
- **Alternative:** Octonion multiplication is alternative, which is weaker than associativity but still ensures some structure.
- **Normed division algebra:** The norm of an octonion  $O = (q_1, q_2)$  is defined as:

$$N(O) = N(q_1) + N(q_2)$$

where  $N(q_1)$  is the norm of the quaternion  $q_1$ . The octonions form a normed division algebra, meaning you can divide one octonion by another (except by zero).

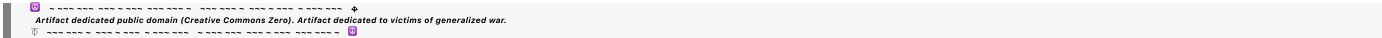
## Summary of Cayley-Dickson Rules

- The construction doubles the dimension at each step: real numbers  $\mathbb{R}$  (1D) to complex numbers  $\mathbb{C}$  (2D), complex numbers to quaternions  $\mathbb{H}$  (4D), and quaternions to octonions  $\mathbb{O}$  (8D).
- The multiplication rule for octonions is defined in terms of quaternion pairs, with non-associativity emerging at the octonion stage.

This construction generalizes further to sedenions (16D) but loses properties like normed division in higher dimensions.

Let me know if you'd like any further breakdowns or visual aids!

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